

Final Exam DSC 105 (18-Dec-2025)

1. What command checks the structure of a built-in dataset like `mtcars` in R?
`str(mtcars)` (shows 32 observations and 11 variables). What other information is contained in that output?
2. How do you compute the base-10 logarithm of 10 in R?
`log(10, base=10)` or `log10(10)` (returns 1).
3. What are R's special values for infinity, not-a-number, and not-available?
Infinity: `Inf` or `-Inf`; Not-a-number: `NaN`; Not-available: `NA`.
4. How do you test for near equality in R due to floating-point precision issues, and give an example using `sqrt(2)2` and 2?
Use `all.equal(sqrt(2)2, 2)` (returns `TRUE` for near equality); `identical(sqrt(2)2, 2)` returns `FALSE` due to rounding errors.
5. What do you get when applying the 'summary' function on a categorical variable?
6. What is vectorization in R, and give an example.
Applies operations element-wise; `c(1, 2, 3) * 2` returns `c(2, 4, 6)`.
7. What is a time series? Do you know of an example? What plot type is appropriate?
A time series is a sequence of data points collected or recorded at successive, equally spaced time intervals, often to track changes over time. It differs from cross-sectional data by emphasizing chronological order and temporal dependencies. The appropriate plot is a line plot for time as the dependent variable, or a histogram to plot frequencies (counts) across the dataset.
8. Explain what you see when you make a boxplot, and which data can be used to make one.
Same as summary without the mean, median, 1st, 3rd quartile (IQR), outliers, box and whiskers.
9. Give a basic `ggplot2` example to plot `mpg` vs. `wt` from `mtcars` as a scatterplot.
`library(ggplot2); ggplot(mtcars, aes(wt, mpg)) + geom_point()`.